
The Processing of Expletive Negation in Italian Temporal Clauses: A Self-Paced Reading study

Anna Teresa Porrini^{1*}, Veronica D'Alesio^{1*}, and Matteo Greco¹

¹University School for Advanced Studies - IUSS Pavia

Abstract:

This work aims to investigate the role of negation when it fails to reverse the polarity of a sentence, thus becoming 'expletive'. An example is offered by the following Italian sentence: *Chiara è rimasta in casa finché Marco **non** ha chiamato la pizzeria per la cena* (lit. 'Chiara remained in the house until Marco did **not** call the pizzeria for dinner' = 'Chiara remained at home until Marco called the pizzeria for dinner'). Although negation occurs, the sentence remains affirmative and the negative marker fails to trigger all the semantic and syntactic phenomena usually associated with it. Here we implement two experiments, an Acceptability judgment task and a Self-Paced Reading paradigm, to test how expletive negation is understood and processed. Our findings reveal a complex picture of expletive negation in Italian. Acceptability judgments suggest that it increases sentence naturalness. The self-paced reading data show no additional processing cost associated with expletive negation, unlike standard negation. By integrating offline and online measures, our study represents a crucial step toward a more comprehensive understanding of expletive negation in linguistic processing, which seems to diverge from the processing of standard negation.

Keywords: Negation; Expletive negation; Acceptability judgment; Self-Paced Reading; Sentence processing.

*Corresponding author: anna.porrini@iusspavia.it

*Joint first authors

1 Introduction

Languages do not always appear as logical as they might be. For example, they may include negative markers that do not reverse the truth conditions associated with a proposition, which remains affirmative (Jespersen 1917: 75). Consider the following example in Italian:

- (1) *Rimarrò in stazione finché non arriva il mio amico*
 stay.1stsing.fut in station until NEG arrives the my friend
 ‘I will be staying at the (train/bus) station until my friend arrives’

In this sentence, the occurrence of the Italian negative morpheme “non” does not reverse the polarity of the proposition, exemplifying the phenomenon of ‘Expletive Negation’ (EN). Conversely, the single negation interpretation of a sentence containing a negative marker that does reverse its polarity is referred to as ‘Standard Negation’ (SN).

EN occurs in a limited number of languages – 74 out of a survey of 722 languages (Jin & Koenig 2021: 40) – and has garnered increasing interest over the past few decades (Horn 1989, 2010; Espinal 1992; Del Prete 2008; Yoon 2011; Makri 2013; Greco 2019a, 2020; Jin & Koenig 2021, Ramat 2022). It is worth noting that languages displaying EN show various instances of it, making this phenomenon highly productive. In the case of Italian, for instance, according to Greco (2019a), there are at least 11 types of EN, which can be found both in main and subordinate clauses. Among the different types of EN sentences, some received less attention in the experimental literature, such as temporal clauses as the one displayed in (1) (see Martínez 2023, 2024 and the references therein). The present study attempts to close this gap.

EN has been analysed in literature as either a non-negative element or a negative one. According to the former account, the negative morpheme is either semantically empty (Fischer 1992; Van der Wurff 1999; Espinal 1992, van der Wouden 1994a, 1994b, among others) or an evaluative mood marker (Yoon 2008, 2011)/epistemic modal marker (Makri 2013). In the sentence in (1), the emptiness of negation would be highlighted by the fact that the sentence remains affirmative and the negative morpheme is optional. According to the accounts that consider EN a negative element, the negative morpheme always realises a negative operator, but it does not operate on a propositional layer. Therefore, it does not affect the truth-value conditions associated with a sentence (Abels 2005; Greco 2019b, 2020; Delfitto et al. 2019, a.o.). For example, EN could highlight that the speaker in (1) will remain at the (train/bus) station for the entire duration preceding the friend's arrival – that is, for the entire period during which the friend has **not** yet arrived, an account similar to the one proposed for other instances of EN presented in exclamative sentences (Portner & Zanuttini 2000).

1.1 Literature review

Although many studies have focused on the elaboration of SN in natural languages, very little research has been conducted on EN. Concerning the former, it has been observed that negative sentences take longer time to be processed and result in higher error rates compared to non-negative sentences, regardless of the experimental task (Wason 1965; Clark and Chase 1972; Carpenter and Just 1975; Trabasso et al. 1971; Carpenter et al. 1999; Hasegawa et al. 2002; Kaup et al. 2007). Moreover, it has also been shown that negation is integrated into the discourse model only at a later stage, sometimes even after several seconds following the end of the sentence in which it appears (Hasson and Glucksberg 2006; Giora 2006; Kaup et al. 2005, 2006; Lüdtke et al. 2008; Scappini et al. 2015). The increased processing time, higher error rates, and delayed integration have been interpreted as outcomes of two principles: negative sentences necessitate more processing effort than their affirmative counterparts (Carpenter & Just 1975; Carpenter et al. 1999; Hasegawa et al. 2002; Dale & Duran 2011), and linguistic negation diminishes the accessibility of information within its scope (MacDonald & Just 1989; Kaup 2001; Tettamanti et al. 2008; Liuzza et al. 2011; Bartoli et al. 2013). These two principles remain a subject of ongoing debate, yet there is general agreement regarding the effect of negation's occurrence.

Coming to EN, very few experimental studies have addressed this topic (see, for example, Tubau et al. 2017). Among them, Greco et al. (2020) focused on a specific Italian EN clause – namely the Surprise Negation sentence (Snegs), illustrated by the following sentence:

- (2) *E* **non** *mi* *è scesa dal treno Maria?!*
 and **not** CLITIC.to_me is got off-the train Mary'
 'Maria got off the train (and that was surprising)'

Snegs occur in specific contexts where speakers are reacting to unexpected events (hence the label "Surprise") and wish to express this reaction. In their visual-world eye-tracking experiment, Greco et al. (2020) showed that EN clauses caused fixation patterns similar to the ones caused by affirmative clauses, while striking differences were observed between standard negative and affirmative sentences. Interestingly, they observed that negation has an immediate effect on sentence processing, occurring much earlier than what was expected from the literature (Hasson and Glucksberg 2006; Giora 2006; Kaup et al. 2005, 2006; Lüdtke et al. 2008; Scappini et al. 2015, a.o.). Additionally, results indicated that Snegs demand greater processing effort compared to affirmative sentences, as reflected in slower response times and lower accuracy. Their findings suggest that Snegs are mainly affirmative, as negation does not affect their predicative core due to their specific syntactic configuration, but exhibit a more complex pragmatic integration into the discourse (see Greco [2019b] for a detailed theoretical discussion).

1.2 The present studies

The present studies aimed to investigate EN and to describe its behaviour in offline comprehension and online processing. More specifically, the main research question was to ascertain whether EN can be considered an empty element. Since the integration of SN has proved to be challenging in terms of cognitive effort, we believe that a study exploring EN should adopt a similar approach and be able to determine any possible cognitive cost associated with its processing. Our research design thus involved two experiments. First, as detailed in Section 2, an acceptability judgment task was conducted to address the naturalness and the comprehensibility of such structures in Italian. The results show that, despite its optionality, the presence of an expletive negative marker is considered more natural than its absence in temporal until-clauses in Italian. Then, in Section 3 we describe a separate self-paced reading task (henceforth, SPR) targeting the online sentence processing and comprehension of the same stimuli. We found that EN does not lead to higher error rates in comprehension questions and it does not entail additional cognitive cost when reading (contrary to what happens for SN). The findings from both experiments are discussed in Section 4. Both studies were approved by the Ethics Committee of the University of Pavia (prot. 27).

The full lists of stimuli, datasets and full analysis script for both experiments are available online on OSF: <https://osf.io/d6vf7/>.

2 Experiment I

Our first experiment was an acceptability Judgment task, aimed at assessing the comprehensibility and the naturalness of subordinate temporal clauses containing EN (as in (1)). The presence of EN is optional in these clauses, so that the propositional meaning of the sentence remains the same whether the element ‘non’ is there or not (Cépeda, 2017). While arguing for its semantic emptiness is in line with a surface interpretation of the phenomenon, such a claim would not account for a general principle of economy (Zipf, 1949; Martinet, 1960) or efficiency (Hawkins, 2004). Therefore, our first research question focused on whether a subordinate clause containing EN is more or less acceptable than its affirmative counterpart. Given their equal propositional meaning, if EN is a vacuum element, it should be less preferable than its affirmative counterpart, which is a simpler structure in terms of number of elements. On the contrary, if EN is a negative marker, it should be equally or more acceptable than its corresponding affirmative alternative, as its presence would be needed to accomplish a specific function in the sentence. SN and a corresponding affirmative clause were also tested within the same experiment. We hypothesized SN and affirmative clauses to be equally acceptable in terms of comprehensibility and naturalness, as they each deliver their own meaning.

2.1 Participants

Eighty adult native speakers of Italian (age range: 18-72; mean age: 32; 59 females, 19 males and 2 who preferred not to say), with no self-reported learning and/or language disorders, took part in the experiment. All participants provided informed consent and participated voluntarily to the experiment, with no contribution. Pre-screening occurred as participants who confirmed that Italian was their first language were taken into consideration.

2.2 Materials and procedure

The experimental materials consisted of a total of 88 randomized trials, of which 32 critical trials and 66 filler trials, comprising 4 practice trials and 16 items with manipulations unrelated to the present study. The number of critical items per condition (as well as the number of participants) was established following indications by Keating and Jegerski (2015). Critical trials were arranged in a 2x2 factorial design, presenting sentence type (causal/temporal) and negation status (i.e. absence or presence of the negative marker *non*, leading to affirmative/negative sentences) as independent variables, thus resulting in 4 counterbalanced lists. The sentence type variable allowed for the manipulation of both SN and EN within the same study. As an instance of EN, we analysed temporal subordinate clauses introduced by *finché* (Karttunen 1974; Condoravdi 2002; Giannakidou 2002), while for SN, we analysed causative subordinate clauses introduced by *perché* (Rizzi 2001).

To ensure the suitability of the stimuli for a future reading paradigm as well, the presence of the negative particle *non* was balanced out in the affirmative counterparts by lengthening the proper noun by one syllable (e.g., *Marco* and *Marcello*). Example sentences for the four experimental conditions (Temporal Negative, TN; Temporal Affirmative, TA; Causal Negative, CN; Causal Affirmative, CA) are provided in Table 1.

The until construction involved a main clause containing a durative predicate (e.g., to remain) and a temporal clause specifying the conclusion, or endpoint (e.g., to call the pizza restaurant), of the situation described in the main clause (Karttunen 1974; Kortmann 1997). This type of until clause is often labelled as “durative” (see Karttunen 1974, Giannakidou 2002; Macdonald and Ürögdi 2011; Wälchli 2018 for an overview of the different types of until-clauses in several languages). In fact, the temporal conjunction *finché* can have delimitative (“until”) or coextensive (“as long as”) interpretation. In our critical items, the non-durative verbs chosen ensured that the delimitative reading is the only possible one. More specifically, in the matrix clause, the use of the stative verb “rimanere” (to remain) in perfective tense (“è rimasto”) constrained the possible interpretations of the negative marker after the temporal conjunction, making expletive negation the only option (Delfitto 2020), whereas in the subordinate clause, the presence of achievement verbs in perfective tense (“ha chiamato”) disambiguated between the two possible meanings of *finché* in Italian.

Tab. 1: Example of critical items sorted for condition and divided by chunks.

Temporal Negative (TN)						
Chiara è rimasta	in casa	finché	Marco non	ha chiamato	la pizzeria	per la cena
Chiara stay _{3rd.sing.fem.past}	in house	until	Marco neg	call _{3rd.sing.past}	the pizzeria	for the dinner
‘Chiara remained in the house until Marco called the pizzeria for dinner’						
Temporal Affirmative (TA)						
Chiara è rimasta	in casa	finché	Marcello	ha chiamato	la pizzeria	per la cena
Chiara stay _{3rd.sing.fem.past}	in house	until	Marcello	call _{3rd.sing.past}	the pizzeria	for the dinner
‘Chiara remained in the house until Marcello called the pizzeria for dinner’						
Causal Negative (CN)						
Chiara è rimasta	in casa	perché	Marco non	ha chiamato	la pizzeria	per la cena
Chiara stay _{3rd.sing.fem.past}	in house	because	Marco neg	call _{3rd.sing.past}	the pizzeria	for the dinner
‘Chiara remained in the house because Marco did not call the pizzeria for dinner’						
Causal Affirmative (CA)						
Chiara è rimasta	in casa	perché	Marcello	ha chiamato	la pizzeria	per la cena
Chiara stay _{3rd.sing.fem.past}	in house	because	Marcello	call _{3rd.sing.past}	the pizzeria	for the dinner
‘Chiara remained in the house because Marcello called the pizzeria for dinner’						

The experiment was carried out online on PennController for Internet Based Experiments (PCIBex, Zehr and Schwarz 2018). Participants were recruited through social media and provided informed consent before being asked to read each sentence and then rate its naturalness and comprehensibility on a 7-point Likert scale. To ensure the accuracy of the task, filler items presented varying degrees of acceptability (high, medium, low). An example of a filler item as presented to the participant is pictured in Figure 1.

Fig. 1. Example of a filler sentence (‘To fall asleep faster I drink a chamomile tea with Elsa every night’), followed by the question ‘How natural and understandable does this sentence sound to you?’ and a 7-point Likert scale going from 1 ‘not at all’ to 7 ‘completely’.

Per addormentarmi più velocemente bevo una camomilla con Elsa tutte le sere.

Quanto ti sembra naturale e comprensibile questa frase?

(per niente) ○ ○ ○ ○ ○ ○ ○ (del tutto)

Continua

2.3 Data analysis

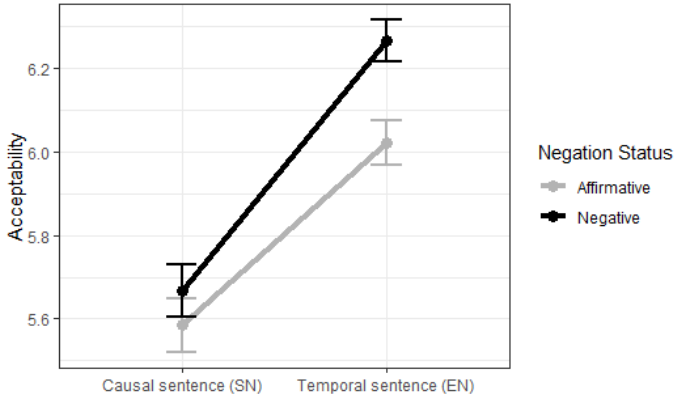
All participants correctly rated filler sentences as scarcely, averagely, or highly acceptable. Data analysis for the acceptability ratings was performed with logistic ordinal regression (clmm) with logit link using the ordinal package in R (Christiansen, 2023). The p-values for the fixed effects were obtained using Wald z-tests, as implemented in the clmm() function.

We included sentence type (causal vs. temporal) and negation status of the sentence (affirmative vs. negative), as well as their interaction, as fixed factors. Prior to analysis, sentence type and negation status were contrast coded: temporal sentence type corresponded to -1 while causal to +1; similarly, for negation status negative sentences were coded as -1 and affirmative ones as +1. The random effects structure was established in a data-driven manner, beginning with the maximal random effects model (following Barr et al., 2014).

2.4 Results

Acceptability ratings for critical items in each condition can be seen in Figure 2 and Table 2. The results show significant effects of sentence type ($\beta = -0.50$, $SE = 0.08$, $z = -6.30$, $p < .001$) and negation status ($\beta = -0.24$, $SE = 0.07$, $z = -3.35$, $p < .001$), as well as a significant interaction between the two ($\beta = 0.16$, $SE = 0.04$, $z = 3.80$, $p < .001$). Post-hoc contrast analyses (emmeans) indicated that all contrasts between the different conditions were significant, except for the contrast between the negative and affirmative version of the causal sentence type. Tables summarising the results of all analyses can be found in the Supplementary Materials.

Fig. 2: Mean acceptability ratings in the norming study for each of the four experimental conditions, with error bars calculated based on standard errors.



Tab. 2: Mean acceptability, standard error and median acceptability for each of the four experimental conditions.

Condition	Mean	Standard error	Median
Temporal Negative (TN)	6.27	0.05	7
Temporal Affirmative (TA)	6.02	0.05	7
Causal Negative (CN)	5.67	0.06	6
Causal Affirmative (CA)	5.58	0.06	6

2.5 Interim discussion

All conditions showed high acceptability (i.e. above 5). As expected, negative and affirmative causal sentences were equally acceptable for adult native speakers of Italian. However, causal sentences were generally considered less acceptable than temporal ones. This may be attributed to the fact that our critical sentences needed to convey a coherent meaning in both their affirmative and negative form. Therefore, for causal sentences the event described in the matrix clause was not always a direct consequence of the event expressed in the subordinate clause, which in some instances could have made the interpretation of the causal relation less evident. Interestingly, in temporal clauses the negative condition displaying EN was deemed significantly more comprehensible and natural than the affirmative one. Participants therefore showed a preference for temporal sentences containing a negative particle, even though it did not alter its propositional meaning. It is likely that its contribution in terms of efficiency is shown at a non-surface level, as already suggested for temporal clauses by Delfitto (2020) for EN in before-clauses and by Cèpeda (2017) for until-clauses.

3 Experiment II

Experiment I revealed a preference for EN in temporal clauses for adult native speakers of Italian. These results seem to argue against the semantic emptiness of EN. To further explore this hypothesis, we conducted a second experiment involving an online methodology, SPR, that takes into account the processing of EN while reading. Sentences were presented in separate segments and participants were instructed to press a button to reveal the following chunk, while the previous one disappeared. In this way, participants controlled the rate at which they read, making comprehension self-paced. The time taken to read the segment and press the button (i.e., reaction times, RTs) is considered to be reflective of cognitive effort (Jegerski, 2013; Keating & Jegerski, 2015). Consequently, we expected causal sentences containing SN to show longer RTs compared to their affirmative counterparts, in line with previous research observing difficulties associated with the processing of SN (a.o., Kaup et al., 2007). More precisely, we expected differences to emerge on the verb under scope of negation. If EN is indeed a negative marker, we expected the same behaviour for temporal clauses. However, since EN shares the same

propositional content of an affirmative clause, the presence of the negative marker *non* in this condition (TN) compared to its affirmative counterpart (TA) may lead to a smaller negation effect (i.e., a smaller magnitude of the difference between levels of negation status) than what is observed for SN comparing CN to CA. This picture would be in line with the observations made on Snegs by Greco et al. (2020), and would ascribe expletive temporal negative clauses to a class of negative sentences whose surface meaning is mainly affirmative. As a secondary task, we asked participants comprehension questions, serving a twofold purpose: first, we ascertained that negative and affirmative temporal sentences were understood as bearing the same propositional content; secondly, we used these data as an accuracy measure. In line with previous studies (for a review, see Kaup and Dudschig, 2020), we expected questions following SN to entail higher error rates compared to questions following affirmative sentences. Similarly, if EN works as a negative marker, the same should be true for questions following it.

3.1 Participants

Eighty adult native speakers of Italian (age range: 21-45; mean age: 30; 43 females, 37 males), with no self-reported learning and/or language disorders, participated in the experiment. Participants were not pre-screened within the experiment. However, the following requirements were flagged among the options available to identify our target population: first language (Italian), main language (Italian), age (18-45), country of residence (Italy). All participants provided informed consent and were rewarded 1.90€ for their participation.

3.2 Materials and Procedure

The same stimuli as Experiment I were used, except for 4 items that presented low acceptability scores (Q1 below 5) in the previous judgments and were therefore excluded from further investigation. Consequently, the SPR task contained 28 randomized critical trials. The task was implemented in PennController for Internet Based Experiments (PCIBex, Zehr and Schwarz 2018) and then administered via Prolific, an online participant recruitment platform. Sentences were presented in a masked chunk-by-chunk non-cumulative SPR paradigm. Chunks were preferred to single-word presentation in order to compensate for the lack of the negative particle in the affirmative conditions. They were kept between 1 and 3 words and defined by the authors as follows:

- Chunk 1: a proper name {Chiara};
- Chunk 2: the stative verb “to remain” in perfective tense {è *rimasta*, remained};
- Chunk 3: a locative adverbial of place {*in casa*, at home};
- Chunk 4: the causal or temporal conjunction (i.e., sentence type manipulation) {perché/finché, *because/until*};

- Chunk 5: a proper noun and, potentially, the negative marker “non” (i.e., negation status manipulation) {Marcello/Marco non, *Marcello/Marco did not*};
- Chunk 6: an achievement, non-durative verb in perfective tense {ha chiamato, *called/call*};
- Chunk 7: the object of the verb, composed of article + noun {la pizzeria, *the pizzeria*};
- Chunk 8: a final prepositional phrase {per la cena, *for dinner*}.

Participants were instructed to press the spacebar on their keyboard to reveal the following chunks and to move on to the following item. Comprehension questions followed each item and the ratio between “yes” and “no” answers was kept equal. Questions were visible all at once, along with “1. yes” and “2.no” buttons signalled on the screen. Participants could either click the pointer cursor on the relevant button or press “1” for “yes” and “2” for “no” on their keyboard. The entire trial is displayed in Figure 3. Four initial practice trials were presented in order to let the participants become familiar with the apparatus and the procedure. In this stage, written feedback was provided after each response following comprehension questions (either ‘Wrong answer, read more carefully!’ or ‘Correct answer!’), while during the experimental phase no feedback was provided.

Fig. 3. Example of a critical trial from beginning to end. The sentence and the question pictured here correspond to condition CA (Answer: YES) in Table 1 and 2 respectively.

Chiara	_____	_____	_____	_____	_____	_____	_____	_____	_____
_____	è rimasta	_____	_____	_____	_____	_____	_____	_____	_____
_____	_____	in casa	_____	_____	_____	_____	_____	_____	_____
_____	_____	_____	perché	_____	_____	_____	_____	_____	_____
_____	_____	_____	_____	Marcello	_____	_____	_____	_____	_____
_____	_____	_____	_____	_____	ha chiamato	_____	_____	_____	_____
_____	_____	_____	_____	_____	_____	la pizzeria.	_____	_____	_____

Chiara è rimasta in casa?

1. sì

2. no

Participants were given the possibility of a mid-experiment break but were asked to go through each half in one sitting. The task lasted 16 minutes on average.

Questions following filler items addressed the general meaning of the previous sentence. Two questions were designed for each critical item, with either “yes” or “no” as an answer. This was meant to ensure that questions following negative critical items could

be specifically targeted to assess the comprehension of the subordinate clause (i.e., whether the event in the clause is negated by the marker *non*), while avoiding an overlap between our experimental conditions and the “yes” or “no” answers associated with EN and SN respectively. In order to balance “yes” and “no” responses, 8 counterbalanced lists were created. Examples of the questions in the four conditions are reported in Table 3, referring to the critical items displayed in Table 1.

Tab. 3: Example of questions following critical items in each condition

Condition	Sentence	Answer
TN	<i>Marco ha chiamato la pizzeria per la cena?</i>	<i>Sì</i>
	‘Did Marco call the pizzeria for dinner?’	YES
	<i>È Chiara che ha chiamato la pizzeria per la cena?</i>	<i>No</i>
	‘Was it Chiara who called the pizzeria for dinner?’	NO
CN	<i>Chiara è rimasta in casa?</i>	<i>Sì</i>
	‘Did Chiara remain in the house?’	YES
	<i>Marco ha chiamato la pizzeria per la cena?</i>	<i>No</i>
	‘Did Marco call the pizzeria for dinner?’	NO
TA	<i>Marcello ha chiamato la pizzeria per la cena?</i>	<i>Sì</i>
	‘Did Marcello call the pizzeria for dinner?’	YES
	<i>È Chiara che ha chiamato la pizzeria per la cena?</i>	<i>No</i>
	‘Was it Chiara who called the pizzeria for dinner?’	NO
CA	<i>Chiara è rimasta in casa?</i>	<i>Sì</i>
	‘Did Chiara remain in the house?’	YES
	<i>È Chiara che ha chiamato la pizzeria per la cena?</i>	<i>No</i>
	‘Was it Chiara who called the pizzeria for dinner?’	NO

3.3 Data analysis

No participant was excluded due to low accuracy (< 75%) in their answers to the filler item questions. Data analysis was performed on the accuracy of the answers given to comprehension questions following critical items and on the RTs of Chunks 5 to 8 of those same items. The full analysis script for both experiments, comprising all model outputs, are available as supplementary materials.

3.3.1 Comprehension Questions

The comprehension questions following the experimental items were all yes/no questions with unambiguously correct answers. The participants' answers were therefore coded as correct (1) or incorrect (0). The statistical analysis for accuracy was performed with a generalised linear mixed model (glmm) assuming a binomial distribution, using the lme4 package in R (Bates et al., 2015). P-values were obtained as in Experiment I. Sentence type, negation status and their interaction were included as fixed factors in the model, coded as in Experiment I. The random effects structure was determined in a data-driven manner, starting with the maximal random effects model. The random structure was then simplified by systematically removing random effects one at a time, guided by the results of a randomised principal component analysis (rePCA), until the issue of singular fit was resolved.

3.3.2 Reaction Times

The data trimming procedures were the following: first, only RTs for trials followed by correct answers to the comprehension question were considered for statistical analysis. This choice was made a priori, considering how in trials with incorrect answers it could not be assumed that the participant had paid attention to the stimulus. Given the fact that the average overall accuracy was above 95%, this exclusion criterion resulted in no significant data loss. Then, RTs under 100 milliseconds and over 3000 milliseconds were excluded. These were established as fixed a priori thresholds: the former is a minimal physiological limit for reacting to a visual stimulus, while the latter is meant to remove potential artefacts due to participants' distraction during the online task. No further exclusion of outliers was performed (following Miller, 2023). Following data trimming, a logarithmic transformation was applied to the RTs to normalise the data distribution, in line with the approach to linguistic RT data proposed by Baayen (2008).

To investigate the effects of sentence type and negation status on reading times, we fit linear mixed-effects models for Chunks 5 to 8 using the lme4 package in R (Bates et al., 2015), with p-values based on Satterthwaite t-tests. The log-transformed RTs were analysed as a dependent variable, while sentence type, negation status (contrast coded as above) and the interaction between the two constituted the fixed structure of the lme4 models. To fully interpret the interaction, we used treatment coding for the factor sentence type and re-leveled the factor twice, first setting causal sentences as the reference level, then temporal ones. This re-leveling strategy allowed us to estimate and interpret the effect of negation status within each sentence type separately. In each model, the coefficient for negation status reflects its effect within the reference level of sentence type. The random effects structure for each model was determined in a data-driven manner, as above.

3.4 Results

The tables with the full output of the analyses performed for Experiment II, including the full description of each model, can be found in the Supplementary Materials.

3.4.1 Comprehension Question

Descriptive statistics for accuracy on the comprehension questions are shown in Table 4.

Tab. 4: Mean Accuracy and standard error on the comprehension questions.

Condition	Accuracy	Standard error
Temporal Negative (TN)	0.97	0.01
Temporal Affirmative (TA)	0.95	0.01
Causal Negative (CN)	0.94	0.02
Causal Affirmative (CA)	0.97	0.01

The results of the glmm showed no significant effect of either sentence type or negation status, but an interaction between the two ($\beta = 0.33$, $SE = 0.11$, $z = 3.00$, $p = .003$). Post-hoc contrast analyses (emmeans) indicated that the only significant contrast in accuracy is between the CN and CA conditions ($\beta = -0.86$, $SE = 0.31$, $z\text{-ratio} = -2.72$, $p = .033$), while no other contrasts were significant.

3.4.2 Reaction Times

Descriptive statistics for RTs in the analysed chunks are shown in Table 5.

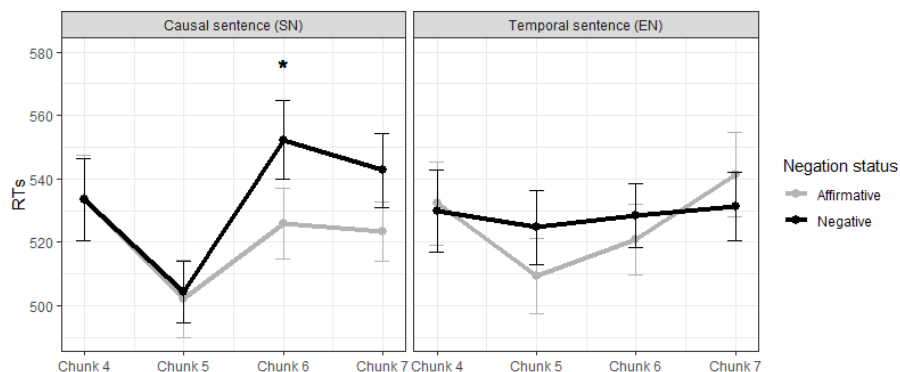
Tab. 5: Mean Reaction times in milliseconds and standard errors (in parentheses) in milliseconds for Chunk 5 (Marcello/Marco did not), Chunk 6 (called/call), Chunk 7 (the pizzeria), and Chunk 8 (for dinner) in the four experimental conditions

Condition	Chunk 5	Chunk 6	Chunk 7	Chunk 8
Temporal Negative (TN)	525 (12)	528 (10)	531 (11)	747 (18)
Temporal Affirmative (TA)	509(12)	521 (11)	541 (13)	730 (18)
Causal Negative (CN)	504 (10)	552 (12)	543 (12)	784 (21)
Causal Affirmative (CA)	502 (12)	526 (11)	523 (9)	774 (19)

On Chunk 5, which is the point at which the negative marker is presented, no significant effect was detected. On Chunk 6, however, a negative effect of negation status was seen when causal sentences (presenting SN) are used as a reference level ($\beta = -0.02$, $SE = 0.01$, $t = -2.11$, $p = .036$). In causal sentences, then, the verb of the subordinate clause is read faster when there is no negative marker preceding it. The same effect was not replicated for temporal sentences ($\beta = -0.02$, $SE = 0.01$, $t = -1.36$, $p = .177$). No significant effects

were found in either model on Chunk 7, while on Chunk 8 there was an effect of sentence type, mirrored in both models ($\beta = |0.04|$, $SE = 0.02$, $t = |2.29|$, $p = .025$), signalling that the final chunk was read faster for temporal sentences. This effect will, however, not be discussed further, as the comparison between sentence types was not of concern for the research questions of the study. A visual representation of RTs for Chunks 4 to 7 in each of the experimental conditions can be seen in Figure 4.

Fig. 4: Mean RTs for Chunk 4 (because/until), Chunk 5 (Marcello/Marco did not), Chunk 6 (called/call), Chunk 7 (the pizzeria), and Chunk 8 (for dinner) in the four experimental conditions, with error bars calculated based on standard errors. The * indicates a significant difference ($p < 0.05$) between the affirmative and the negative condition according to the fitted model.



3.5 Interim discussion

As expected, our results on SN replicated previous findings: the reaction time data confirmed that sentences containing SN are more difficult to process than their affirmative counterparts, consistent with several studies using various experimental paradigms (a.o., Trabasso et al. 1971). Furthermore, the facilitatory effect of affirmative causal clauses compared to their negative counterparts was evident, as predicted, on the chunk containing the verb under the scope of negation. It is interesting to note that the presence of the negative marker alone did not cause an immediate increase in terms of processing effort, as was clear from the analyses on Chunk 5, and a significant negation effect appeared on the negated predicate instead. On the other hand, EN did not show a comparable effect: contrary to what was expected, its presence did not entail a significantly more effortful reading compared to affirmative sentences, neither on the chunk containing the negative marker (Chunk 5), nor on the following verb (Chunk 6). Our results, therefore, highlighted a crucial difference in the processing of EN compared to SN, with the former behaving similarly to an affirmative sentence, as could be inferred by their shared propositional content.

The results of the analyses on comprehension questions were consistent with those of reaction times: they showed lower accuracy after SN, while participants' responses on questions following EN were as accurate as after the affirmative conditions, signalling that EN led to a smaller misinterpretation effect than SN. This outcome replicated previous findings showing lower accuracy after standard negation (for a review: Kaup & Dudschig, 2020) and extended those findings by showing that expletive negation did not generate the same effect.

Interestingly, it is likely that accuracy data on SN does not reflect a proper misinterpretation of the negative sentence by adult native speakers, but rather the mere consequence of a reduced accessibility to the negated information, as observed in various studies (MacDonald and Just 1989; Kaup 2001; Tettamanti et al. 2008; Liuzza et al. 2011; Bartoli et al. 2013; Papitto et al. 2021). In our experimental design, answering comprehension questions once the critical item was gone required retrieving from memory the event stated in the subordinate clause. Therefore, a less accessible element might arguably have caused a higher level of uncertainty when probed. As we have seen from the results, this did not happen for EN, possibly signalling that the two propositions contained in the affirmative and negative temporal sentences were equally interpreted as well as equally accessible. While this is in line with EN not changing the propositional content of the clause, it seems to argue against its nature as a negative marker. It is worth noting, however, that the type of questions provided in this experiment only probed the surface meaning of the sentence. Future research may also address the implicated meaning underlying EN in these types of clauses, which could, in turn, be more prone to lower accuracy rates. The presence of the negative marker *non* could be required by the complex semantics of *finché*, which intertwines with the verb aspect and the temporal relation that it establishes between the matrix and the subordinate clause.

Taken together, results from Experiment II seem to cast doubt on the effective contribution of EN to until-clauses or, at least, to the representation and processing of their surface meaning.

4 General Discussion and Conclusion

Our study combined online and offline measures to explore the comprehension and processing of EN in Italian. First, an acceptability judgement task revealed that, although both were highly acceptable, temporal clauses containing the expletive particle *non* sounded more natural and comprehensible than the ones without negation. A difference in acceptability was not detected between sentences containing SN and their affirmative counterpart. In the following SPR experiment, we found that in line with previous literature, SN leads to inflated processing costs, an effect that was evident through slower RTs on the verb in the scope of negation, but also to lower accuracy on comprehension questions following the target sentences. Interestingly, neither effect was found for EN.

On the one hand, the results of the acceptability judgment support the hypothesis that EN is not semantically empty. This finding is in line with some prominent theoretical accounts (Cèpeda 2017, 2018; Cèpeda and Déprez 2023; Greco et al. 2020; Greco 2019b; Delfitto et al. 2019), describing EN as a negative marker that does not operate on a propositional layer of the sentence. On the other hand, however, the data from the SPR task suggest that participants did not incur any extra processing cost when presented with sentences containing EN as opposed to their affirmative counterparts, contrary to what happened for SN. Taken together, our findings paint a complex picture of the processing and understanding of EN in Italian. More specifically, while it can be stated with confidence that EN behaves differently from SN, it is less clear whether this is a consequence of its semantic vacuity or merely an effect of the scope of the negative marker, which would not interact with the truth-value conditions of the sentence. An important clue might come from the semantics of *until*, which, as discussed in the introduction, can trigger the activation of a negative meaning on its own. If this is the case, then the optional occurrence of EN works as a scope marker for an already activated negation (see Delfitto, 2013), and might therefore be preferred. Such an explanation would also account for the fact that EN in *until* clauses did not require greater processing effort compared to affirmative sentences. This would support an analysis of expletive negation as a negative marker that retains its negative nature, but whose interpretation is modulated by the syntactic and semantic context in which it appears.

Although the results are compelling, a limitation of the SPR paradigm is that it forces participants to read in a strictly linear manner and prevents rereading. This differs from natural reading, where regressions are common. SPR may therefore be less sensitive to late-stage integration processes, which could be crucial for detecting subtle processing costs associated with EN. To address this limitation, future research could employ eye-tracking methodologies, which provide a more fine-grained measure of reading dynamics. Another relevant limitation of the present design is that it did not account for the entirety of the phenomenon of EN in Italian, which encompasses a variety of forms and types (Greco 2019a). Further research should address this issue, an endeavour that would provide a better understanding of the exact contribution given by expletive negative markers to sentences containing them.

All in all, our study was the first to directly assess the processing cost of EN in reading through online methodologies, while also comparing it systematically with SN and affirmative counterparts. This constitutes an important first step toward a more comprehensive understanding of EN across different linguistic contexts.

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